

My attempt at understanding Seiberg-Witten theory

Douwe Hoekstra

2025/11/21

1 Functional setup of Seiberg-Witten equations

Fix an integer m and a real number $p \in (1, \infty)$ such that

$$m + 2 - \frac{4}{p} > 0.$$

Let A_0 be a smooth Hermitian connection on $\det(\sigma)$ and let $\mathcal{A}_\sigma^{m+1,p}$ be the following space:

$$\mathcal{A}_\sigma^{m+1,p} = \{A = A_0 + ia \mid a \in L^{m+1,p}(T^*M)\}.$$

If we omit the p in notation, we take $p = 2$: $\mathcal{A}_\sigma^{m+1} = \mathcal{A}_\sigma^{m+1,2}$.

Then we define $\mathcal{C}_\sigma^{m+1} = L^{k+1,2}(S^+) \times \mathcal{A}_\sigma^{m+1}$. This is the configuration space we will think about.

Next, the target space is

$$\mathcal{Y}^{m,p} = \mathcal{Y}_\sigma^{m,p} = L^{m,p}(S^-) \oplus L^{m,p}(i \wedge_+^2 T^*M)$$

and again $\mathcal{Y}^m = \mathcal{Y}^{m,2}$.

Finally, we define the gauge group to be

$$\mathcal{G}_\sigma^{m+2,p} = \{\gamma \in L^{m+2,p}(M, \mathbb{C}) \mid |\gamma(x)| = 1, \forall x \in M\}; \quad \mathcal{G}_\sigma^{k+2} = \mathcal{G}_\sigma^{k+2,2}.$$

The Seiberg-Witten equations give a map $\text{SW}_\eta: \mathcal{C}_\sigma^{k+1} \rightarrow \mathcal{Y}_\sigma^k$. Then we have the solution set

$$\mathcal{Z}_\sigma^{k+1}(g, \eta) = \text{SW}_\eta^{-1}(0)$$

and the moduli space

$$\mathcal{M}_\sigma^{k+1} = \mathcal{M}_\sigma^{k+1}(g, \eta) = \mathcal{Z}_\sigma^{k+1} / \mathcal{G}_\sigma^{k+2}.$$

Now fix a closed form $\eta \in L^{k,2}(\wedge^2 T^*M)$ with $k \geq 1$. Then the moduli space $\mathcal{M}_\sigma^{k+1}(g, \eta)$ is a subset of the space

$$\mathcal{B}_\sigma^{k+1} = \mathcal{C}_\sigma^{k+1} / \mathcal{G}_\sigma^{k+2}.$$

2 Hausdorff quotient

For our proof we need to following general result on Sobolev Spaces.

Theorem 2.1 (Theorem 7.3 in [1]). *Assume that s_i, s are natural numbers and $1 \leq p_i \leq p < \infty$ ($i = 1, 2$) are real numbers satisfying*

$$1. \ s_i \geq s$$

$$2. \ s \geq 0$$

$$3. \ s_i - s \geq n \left(\frac{1}{p_i} - \frac{1}{p} \right),$$

$$4. \ s_1 + s_2 - s > n \left(\frac{1}{p_1} + \frac{1}{p_2} - \frac{1}{p} \right).$$

Then the map

$$L^{s_1, p_1} \times L^{s_2, p_2} \rightarrow L^{s, p}, (f, g) \mapsto fg$$

is continuous bilinear.

This theorem implies the following results, which are all proven by checking that the given assumptions satisfy the assumptions of the theorem.

Lemma 2.2. *Let k, ℓ be integers. If $k \geq 3$ and $k \geq \ell$, then the map*

$$L^{k, 2} \times L^{\ell, 2} \rightarrow L^{\ell, 2}, (f, g) \mapsto fg$$

is continuous bilinear.

Lemma 2.3. *Let $p > 2$. Then the map*

$$L^{2, 2} \times L^{2, p} \rightarrow L^{2, 2}, (f, g) \mapsto fg$$

is continuous bilinear.

Lemma 2.4. *Let $1 \leq p \leq 4$ and let $p < q$. Then the map*

$$L^{2, 2} \times L^{1, q} \rightarrow L^{1, p}, (f, g) \mapsto fg$$

is continuous bilinear.

Lemma 2.5. *Let $1 \leq q < p < \infty$. Then the map*

$$L^{2, 2} \times L^p \rightarrow L^q, (f, g) \mapsto fg$$

is continuous bilinear.

Proof. From the Sobolev Embedding Theorem it follows that $L^{2, 2} \hookrightarrow L^r$ for all $1 \leq r < \infty$. The result now follows by the Hölder inequality. $\square$

Proposition 2.6. *Let (ψ_n, A_n) and (μ_n, B_n) be sequences in $\mathcal{C}_\sigma^{k+1}$ converging to (ψ, A) and (μ, B) respectively. Suppose that for each n we have a $\gamma_n \in \mathcal{G}_\sigma^{k+2}$ such that*

$$\gamma_n \cdot (\psi_n, A_n) = (\mu_n, B_n).$$

Then there is a subsequence of (γ_n) which converges to $\gamma \in \mathcal{G}_\sigma^{k+2}$. Moreover, we have

$$\gamma \cdot (\psi, A) = (\mu, B).$$

Proof. Since $\gamma_n \cdot (\psi_n, A_n) = (\mu_n, B_n)$, we have

$$d\gamma_n = \frac{1}{2}\gamma_n(B_n - A_n).$$

We note that the operator $d + d^*: \Omega^{\text{even}} \rightarrow \Omega^{\text{odd}}$ is elliptic and that for functions $(d + d^*)\gamma = d\gamma$.

We make a case distinction between $k = 1$ and $k \geq 2$.

Case $k = 1$: We have $\gamma_n \in L^{3, 2}$ so by Morrey's theorem all $\gamma_n \in C^0$. Then we obtain a bound

$$\|d\gamma_n\|_{L^q} \leq C\|\gamma_n\|_{C^0}\|B_n - A_n\|_{L^q}$$

for all $1 \leq q < \infty$. By ellipticity, it follows that $\gamma_n \in L^{1,q}$ for all $1 \leq q < \infty$. Then by lemma 2.4 we have

$$\|d\gamma_n\|_{L^{1,p}} \leq C\|\gamma_n\|_{L^{1,q}}\|B_n - A_n\|_{L^{2,2}}$$

for $1 \leq p \leq 4$ and $q > p$. So $d\gamma_n \in L^{1,p}$ for $1 \leq p \leq 4$ and so by ellipticity we have $\gamma_n \in L^{2,4}$. We have $\frac{1}{4} - \frac{1}{2} < \frac{1}{6} - \frac{1}{4}$, so by the Rellich-Kondrachov theorem we have a compact embedding $L^{2,4} \hookrightarrow L^{1,6}$. So there is a $L^{1,6}$ -convergent subsequence of γ_n , which we will denote still by γ_n . It converges to some $\gamma \in L^{1,6}$. By Morrey's theorem $L^{1,6} \hookrightarrow C^0$, so this subsequence also converges in the C^0 sense. In particular we see that $|\gamma| = 1$ everywhere.

We have $d\gamma_n \rightarrow d\gamma$ in L^6 . By lemma 2.5 we obtain that $\frac{1}{2}\gamma_n(B_n - A_n) \rightarrow \frac{1}{2}\gamma(B - A)$ in L^6 as well. So $d\gamma = \frac{1}{2}\gamma(B - A)$. Then by lemma 2.4 we see that $\frac{1}{2}\gamma(B - A)$ is in $L^{1,p}$ for $1 \leq p \leq 4$. So by ellipticity, $\gamma \in L^{2,4}$. Then by 2.3 we see that $\frac{1}{2}\gamma(B - A)$ is in $L^{2,2}$, so by ellipticity $\gamma \in L^{3,2}$. So $\gamma \in \mathcal{G}_\sigma^{k+2}$. Finally, we have $\gamma_n\psi_n \rightarrow \gamma\psi$ by lemma 2.2 and so $\gamma \cdot (\psi, A) = (\mu, B)$.

Case $k \geq 2$: Then we have $\gamma_n \in L^{k+2,2}$. By the Rellich-Kondrachov theorem this embeds compactly in $L^{k+1,2}$, so we have a convergent subsequence in $L^{k+1,2}$, which we also denote by γ_n . Then as above, we obtain that $|\gamma| = 1$ everywhere and $d\gamma = \frac{1}{2}\gamma(B - A)$. Then by lemma 2.2 we see that $d\gamma \in L^{k+1,2}$, so by ellipticity $\gamma \in L^{k+2,2}$. So $\gamma \in \mathcal{G}_\sigma^{k+2}$. The final assertion again follows from lemma 2.2. $\square$

Remark 2.7. This proposition implies that the action of $\mathcal{G}_\sigma^{k+2}$ is proper on $\mathcal{C}_\sigma^{k+1}$. For more information on proper group actions see appendix A.

Corollary 2.8. *The action $\mathcal{G}_\sigma^{k+2} \curvearrowright \mathcal{C}_\sigma^{k+1}$ is proper.*

Proof. Let $(\psi_n, A_n) \subset \mathcal{C}_\sigma^{k+1}$ a sequence and $\gamma_n \subset \mathcal{G}_\sigma^{k+2}$ a sequence. Assume that $(\psi_n, A_n) \rightarrow (\psi, A)$ and that $\gamma_n \cdot (\psi_n, A_n) \rightarrow (\mu, B)$. Then defining $(\mu_n, B_n) = \gamma_n \cdot (\psi_n, A_n)$ we see that we satisfy the condition of the theorem, so γ_n has a convergent subsequence. Hence the action is proper, by proposition A.2. $\square$

Corollary 2.9. *The quotient space $\mathcal{B}_\sigma^{k+1} = \mathcal{C}_\sigma^{k+1} / \mathcal{G}_\sigma^{k+2}$ is Hausdorff.*

Proof. By proposition A.6, since the group action is proper, the quotient $\mathcal{B}_\sigma^{k+1} = \mathcal{C}_\sigma^{k+1} / \mathcal{G}_\sigma^{k+2}$ is Hausdorff. $\square$

Corollary 2.10. *The Seiberg-Witten moduli space $\mathcal{M}_\sigma^{k+1}(g, \eta)$ is Hausdorff.*

Proof. The moduli space is a subspace of $\mathcal{B}_\sigma^{k+1}$. $\square$

3 Curvature bounds

Our next goal is to show that the moduli space is compact. For this we need to estimate the different terms in the Seiberg-Witten equations. Obtaining the necessary bounds is the objective of this section.

We start by estimating the norm of the Clifford action γ .

Lemma 3.1. *Let $\alpha \in i\Omega_+^2(M)$. Then we have the pointwise identity*

$$|\gamma_\alpha|^2 = 4|\alpha|^2,$$

using the Frobenius norm.

Proof. This identity holds pointwise, so it suffices to check it in a point $x \in M$. The Frobenius norm is given by $|\gamma_\alpha|^2 = \text{tr}(\gamma_\alpha^* \gamma_\alpha)$. Since α is in $i\Omega_+^2(M)$, we have $\gamma_\alpha^* = -\gamma_\alpha$. There is a

orthonormal basis $\{\eta_0, \eta_1, \eta_2\}$ for $\wedge_+^2 T_x^* M$ given by

$$\begin{aligned}\eta_0 &= \frac{1}{\sqrt{2}} (dx^1 \wedge dx^2 + dx^3 \wedge dx^4) \\ \eta_1 &= \frac{1}{\sqrt{2}} (dx^1 \wedge dx^3 - dx^2 \wedge dx^4) \\ \eta_2 &= \frac{1}{\sqrt{2}} (dx^1 \wedge dx^4 + dx^2 \wedge dx^3),\end{aligned}$$

where (x^i) are normal coordinates at x .

A computation shows that this basis has the property that $\gamma(\eta_k)^2 = -2\text{id}$ and $\gamma(\eta_k)\gamma(\eta_l) = 0$ for $k \neq l$ at x . So if we write $\alpha = \sum_{k=0}^2 \alpha_k \eta_k$, then we have with Clifford multiplication that

$$\gamma(\alpha)^2 = \gamma\left(\sum_k \alpha_k \eta_k\right)^2 = -2\left(\sum_{k=0}^2 \alpha_k^2\right) \text{id}.$$

This identity map lives on a (complex) dimension 2 space, so $\text{tr}(\text{id}) = 2$. Hence the norm $|\gamma(\alpha)|^2 = 4|\alpha|^2$. $\square$

We have the following equation for solutions of the unperturbed Seiberg-Witten equation.

Lemma 3.2. *Let (ψ, A) be a solution to the unperturbed Seiberg-Witten equations on a four-manifold X . Then*

$$\|\nabla(\psi)\|_{L^2}^2 + \frac{1}{4} \langle \kappa \psi, \psi \rangle_{L^2} + \frac{\|\psi\|_{L^4}^4}{4} = 0.$$

Here κ denotes the scalar curvature of X .

Proof. Since (ψ, A) is a solution to the Seiberg-Witten equations, we have $\not{D}\psi = 0$, so by the Weitzenböck formula we have

$$0 = \not{D}^2 \psi = \nabla^* \nabla \psi + \frac{\kappa}{4} \psi + \frac{F_A}{2} \cdot \psi.$$

Using the Seiberg-Witten equations, we can therefore write

$$\begin{aligned}0 &= \nabla^* \nabla \psi + \frac{\kappa}{4} \psi + \frac{1}{2} \left(\bar{\psi} \otimes \psi - \frac{|\psi|^2}{2} \text{id} \right) \psi \\ &= \nabla^* \nabla \psi + \frac{\kappa}{4} \psi + \frac{|\psi|^2}{4} \psi.\end{aligned}$$

Taking the L^2 -inner product with ψ then yields the asserted equality. $\square$

This implies the following result for the non-existence of non-trivial spinors.

Corollary 3.3. *Let X be a compact four-manifold. Let $\kappa_X^- = \max_{x \in X} (0, -\kappa(x))$. Then*

$$\kappa_X^- \|\psi\|_{L^2}^2 \geq \|\psi\|_{L^4}^4.$$

In particular, if X has non-negative scalar curvature, then any solution to the unperturbed Seiberg-Witten equations has trivial spinor field.

Proof. We have for all $x \in X$ that $-\kappa(x) \leq \kappa_X^-$, so $\kappa(x) \geq -\kappa_X^-$. Therefore

$$\langle \kappa \psi, \psi \rangle = \int_X \kappa |\psi|^2 \text{vol}_g \geq -\kappa_X^- \|\psi\|_{L^2}^2.$$

Using the equality from the previous lemma, we obtain

$$\frac{1}{4} \|\psi\|_{L^4}^4 \leq \|\nabla \psi\|_{L^2}^2 + \frac{1}{4} \|\psi\|_{L^4}^4 = -\frac{1}{4} \langle \kappa \psi, \psi \rangle \leq \frac{1}{4} \kappa_X^- \|\psi\|_{L^2}^2.$$

The final assertions follows by noting that for a manifold with non-negative scalar curvature $\kappa_X^- = 0$. $\square$

Next, we will derive a pointwise estimate for the spinor.

Lemma 3.4. *Let X be a compact Riemannian four-manifold. Suppose that (ψ, A) is a solution to the Seiberg-Witten equations. Then for every $x \in X$ we have*

$$|\psi(x)|^2 \leq \max(\max_{y \in X}(4|\eta_+(y)| - \kappa(y)), 0).$$

Proof. Note that this inequality is invariant under gauge transformations. Therefore we may use the fact that every solution to the Seiberg-Witten equations is gauge equivalent to a smooth solution to show this inequality. So assume without loss of generality that (ψ, A) is a smooth solution. Then by the Weitzenböck formula and Seiberg-Witten equations we have

$$0 = \nabla^* \nabla \psi + \frac{\kappa}{4} \psi + \frac{|\psi|^2}{4} \psi - i\eta_+ \cdot \psi.$$

Let x_0 be the point of X where $|\psi(x)|^2$ attains its maximum. Then taking the inner product with ψ and evaluating at x_0 we obtain

$$\langle \nabla^* \nabla \psi(x_0), \psi(x_0) \rangle + \frac{\kappa(x_0)}{4} |\psi(x_0)|^2 + \frac{|\psi(x_0)|^4}{4} + \frac{i}{2} \langle \eta_+(x_0) \cdot \psi(x_0), \psi(x_0) \rangle = 0.$$

Let (E_i) be a local orthonormal frame for TM . Then we have

$$\begin{aligned} - \sum_i E_i^2 \langle \psi(x), \psi(x) \rangle &= - \sum_i E_i (\langle \nabla_{E_i} \psi(x), \psi(x) \rangle + \langle \psi(x), \nabla_{E_i} \psi(x) \rangle) \\ &= - \sum_i (\langle \nabla_{E_i} \nabla_{E_i} \psi(x), \psi(x) \rangle + 2 \langle \nabla_{E_i} \psi(x), \nabla_{E_i} \psi(x) \rangle + \langle \psi(x), \nabla_{E_i} \nabla_{E_i} \psi(x) \rangle), \end{aligned}$$

so

$$\Delta(|\psi(x)|^2) + 2|\nabla_{E_i}(\psi(x))|^2 = 2\text{Re}(\langle \nabla^* \nabla \psi(x), \psi(x) \rangle).$$

Then in a local maximum x_0 we have $\Delta(|\psi(x_0)|^2) \geq 0$. So $\text{Re}(\langle \nabla^* \nabla \psi(x), \psi(x) \rangle) \geq 0$. Therefore we obtain

$$0 \geq \frac{\kappa(x_0)}{4} |\psi(x_0)|^2 + \frac{|\psi(x_0)|^4}{4} + \text{Re}(\frac{i}{2} \langle \eta_+(x_0) \cdot \psi(x_0), \psi(x_0) \rangle).$$

Then the Cauchy-Schwarz inequality gives us a bound on $\text{Re}(\frac{i}{2} \langle \eta_+(x_0) \cdot \psi(x_0), \psi(x_0) \rangle)$, namely at most

$$|\eta_+(x_0)| |\psi(x_0)|^2,$$

where we used lemma 3.1. So we have $|\psi(x_0)| = 0$, in which case $\psi \equiv 0$, or we obtain the bound

$$0 \geq \frac{\kappa(x_0)}{4} + \frac{|\psi(x_0)|^2}{4} - |\eta_+(x_0)|,$$

so

$$|\psi(x_0)|^2 \leq 4|\eta_+(x_0)| - \kappa(x_0),$$

so the asserted inequality follows. $\square$

Corollary 3.5. *Let X be a compact Riemannian four-manifold. Suppose that (ψ, A) is a solution to the Seiberg-Witten equations. Then for every $x \in X$ we have*

$$|F_A^+(x)| \leq \frac{1}{2} \max(\max_{y \in X}(4|\eta_+(y)| - \kappa(y)), 0) + |\eta_+(x)|.$$

Proof. The Seiberg-Witten equation gives $F_A^+ = \bar{\psi} \otimes \psi - \frac{|\psi|^2}{2} \text{id} - i\eta_+$, so $|F_A^+(x)| \leq \frac{1}{2} |\psi(x)|^2 + |\eta_+(x)|$, so the result follows from the previous one. $\square$

We have the following result for n forms on a Riemannian $2n$ -manifold.

Lemma 3.6. *Let (M, g) be a Riemannian manifold of dimension 4. Let $\omega \in \Omega^2(M)$. We may write $\omega = \omega^+ + \omega^-$ where ω^+ is self-dual and ω^- is anti-self-dual. Then we have*

$$\int_M \omega^2 = \|\omega^+\|_{L^2}^2 - \|\omega^-\|_{L^2}^2.$$

Proof. We have

$$\omega^\pm \wedge \omega^\pm = \pm \omega^\pm \wedge \star \omega^\pm = \pm \langle \omega^\pm, \omega^\pm \rangle \text{vol}_g = \pm \|\omega^\pm\|^2 \text{vol}_g,$$

and

$$\begin{aligned} \langle \omega^+, \omega^- \rangle \text{vol}_g &= \langle \omega^-, \omega^+ \rangle \text{vol}_g = \omega^- \wedge \star \omega^+ = \omega^- \wedge \omega^+ \\ &= \omega^+ \wedge \omega^- = -\omega^+ \wedge \star \omega^- = -\langle \omega^+, \omega^- \rangle \text{vol}_g, \end{aligned}$$

so $\omega^\pm \wedge \omega^\mp = 0$. So we have

$$\begin{aligned} \int_M \omega^2 &= \int_M (\omega^+ + \omega^-)^2 \\ &= \int_M (\|\omega^+\|^2 - \|\omega^-\|^2) \text{vol}_g \\ &= (\|\omega^+\|_{L^2}^2 - \|\omega^-\|_{L^2}^2). \end{aligned}$$

□

In the next theorem we use the following notation:

$$\begin{aligned} \kappa_X^- &= \max_{x \in X} (0, -\kappa(x)) \\ \kappa_{X, \eta}^- &= \max(\max_{y \in X} (4|\eta_+(y)| - \kappa(y)), 0). \end{aligned}$$

Theorem 3.7. *Let X be a compact Riemannian manifold. Then there are only finitely many spin^c structures up to isomorphism for X such that the moduli space of solutions to the Seiberg-Witten equations is non-empty and has non-negative formal dimension. For any solution (ψ, A) to the Seiberg-Witten equations at which the formal dimension is non-negative and for any $x \in X$ we have*

$$\begin{aligned} |\psi(x)|^2 &\leq \kappa_{X, \eta}^- \\ \|\nabla(\psi)\|_{L^2}^2 &\leq \left(\frac{\kappa_X^-}{4} + \frac{1}{2} \kappa_{X, \eta}^- + \|\eta_+\|_\infty \right) \kappa_{X, \eta}^- \text{vol}(X) \\ |F_A^+(x)| &\leq \frac{1}{2} \kappa_{X, \eta}^- + |\eta_+(x)| \\ \|F_A^+\|_{L^2}^2 &\leq \frac{(\kappa_{X, \eta}^- + 2|\eta_+(x)|)^2}{4} \text{vol}(X) \\ \|F_A^-\|_{L^2}^2 &\leq \frac{(\kappa_{X, \eta}^- + 2|\eta_+(x)|)^2}{4} \text{vol}(X) - 8\pi^2 \chi(X) - 12\pi^2 \text{sign}(X). \end{aligned}$$

Proof. The first and third inequality were already established above. The fourth one follows from the third one by integrating over X . For the fifth one, we have formal dimension

$$c_1(\mathcal{L})^2 - (2\chi(X) + 3\text{sign}(X)) \geq 0$$

and by lemma 3.6 we have

$$c_1(\mathcal{L})^2 = \frac{1}{4\pi^2} (\|F_A^+\|_{L^2}^2 - \|F_A^-\|_{L^2}^2),$$

Combining these two facts yields the fifth inequality. Finally, the second inequality we obtain from the Weitzenbock formula

$$0 = \nabla^* \nabla \psi + \frac{\kappa}{4} \psi + \frac{F_A^+}{2} \cdot \psi$$

and the bound on $|F_A^+(x)|$ for all $x \in X$.

It remains to show that there are only finitely many spin^c structures up to isomorphism with non-empty moduli space and non-negative formal dimension. Suppose we have a solution (ψ, A) of the Seiberg-Witten equations at which the formal dimension is non-negative. Then by the preceding, we have a bound depending only on the geometry of X on both $\|F_A^+\|_{L^2}^2$ and $\|F_A^-\|_{L^2}^2$. So the cohomology class represented by $\frac{i}{2\pi} F_A$ lies in a compact subset of $H^2(X, \mathbb{R})$. Moreover, since this is the Chern class it must also be integral, so there are only finitely many possibilities for this class. So there are only finitely many spin^c structures whose determinant line bundle has this given first Chern class. $\square$

4 Gauge fixing

We need the following result to show the compactness of the moduli space.

Lemma 4.1 (Gauge-Fixing Lemma). *Let $\mathcal{L}$ be a complex line bundle over X with a hermitian metric. Fix a unitary C^∞ connection A_0 on $\mathcal{L}$. Then for any $\ell \geq 0$ there are constants $K, C > 0$ depending only on X, A_0, ℓ such that the following hold: For any $L^{\ell,2}$ unitary connection A on $\mathcal{L}$ there is an $L^{\ell+1,2}$ change of gauge σ such that $\sigma \cdot A = A_0 + \alpha$ where $\alpha \in L^{\ell,2}(T^*X \otimes i\mathbb{R})$ satisfies $d^* \alpha = 0$ and*

$$\|\alpha\|_{L^{\ell,2}}^2 \leq C \|F_A^+\|_{L^{\ell-1,2}}^2 + K.$$

Proof. Let $a_0 = A - A_0 \in L^{\ell,2}(T^*X \otimes i\mathbb{R})$. By the Hodge decomposition we find $f \in L^{\ell+1,2}(X, i\mathbb{R})$, $\beta \in L^{\ell+1,2}(\wedge^2 T^*X \otimes i\mathbb{R})$ and ω a harmonic 1-form such that

$$a_0 = df + d^* \beta + \omega.$$

Then $\gamma := \exp(\frac{1}{2}f) \in L^{\ell+1,2}$. Then we have $d\gamma = \frac{1}{2}\gamma df$ and we have

$$\gamma \cdot A = A - 2\gamma^{-1}d\gamma = A - df = A_0 + df + d^* \beta + \omega - df = A_0 + d^* \beta + \omega.$$

Define $\alpha = d^* \beta + \omega$. Now, F_A^+ is the self-dual part of the curvature of the connection A , so in particular we have $d^+ A = F_A^+$. Then it follows that

$$(d^* + d^+)(d^* \beta + \omega) = d^+(d^* \beta + \omega) = d^+(\gamma \cdot A - A_0) = F_A^+ - F_{A_0}^+.$$

Then ellipticity of (d^*, d^+) gives us the following bound for a constant C depending only on X and ℓ (here and in the following C may be increased between estimates)

$$\|d^* \beta\|_{L^{\ell,2}}^2 \leq C \|(d^* d^* \beta, d^+ d^* \beta)\|^2 = C \|F_A^+ - F_{A_0}^+\|_{L^{\ell-1,2}}^2 \leq C \|F_A^+\|_{L^{\ell-1,2}}^2 + C \|F_{A_0}^+\|_{L^{\ell-1,2}}^2.$$

Then with $K_1 = \|F_{A_0}^+\|_{L^{\ell-1,2}}^2$ we have

$$\|d^* \beta\|_{L^{\ell,2}}^2 \leq C \|F_A^+\|_{L^{\ell-1,2}}^2 + CK_1.$$

Next, we need to bound the harmonic component ω . We do this by applying a further gauge transformation.

Let H^1 denote the space of purely imaginary harmonic 1-forms. Let $\omega_0 \in H^1$ and assume that all periods of ω_0 lie in $2\pi i\mathbb{Z}$. Let $\tilde{X}$ be the universal cover of X . Then integrating ω_0 along curves starting at a base point $x_0 \in X$ gives a map $f: \tilde{X} \rightarrow i\mathbb{R}$. Then $\tilde{\varphi} = \exp(f)$ descends to a map $\varphi: X \rightarrow S^1$ since all periods lie in $2\pi i\mathbb{Z}$. Then $d\varphi = \varphi d\tilde{f} = \varphi \omega_0$, so $\omega_0 = \varphi^{-1}d\varphi$.

Let Λ denote the lattice of purely imaginary harmonic 1-forms in H^1 with periods in $4\pi i\mathbb{Z}$. Then H^1/Λ is a torus, so there exists a constant K_2 depending on ℓ such that any ω_0 can be written as $\omega_0 = \omega_1 + 2\omega_2$ with $\omega_2 \in \frac{1}{2}\Lambda$ and $\|\omega_1\|_{L^{\ell,2}}^2 \leq K_2$.

We now apply this to our harmonic component $\omega = \omega_1 + 2\omega_2$. Then ω_2 has periods in $2\pi i\mathbb{Z}$, so there exists a $\varphi: X \rightarrow S^1$ such that $\omega_2 = \varphi^{-1}d\varphi$. So we have

$$\varphi \cdot (A_0 + d^*\beta + \omega) = A_0 + d^*\beta + \omega - 2\varphi^{-1}d\varphi = A_0 + d^*\beta + \omega_1.$$

So we for $\alpha = d^*\beta + \omega_1$ and $\sigma = \varphi\gamma$ we have $d^*\alpha = 0$, $\sigma \cdot A = A_0 + \alpha$ and

$$\|\alpha\|_{L^{\ell,2}}^2 \leq \|\omega_1\|_{L^{\ell,2}}^2 + \|d^*\beta\|_{L^{\ell,2}}^2 \leq K_2 + CK_1 + C\|F_A^+\|_{L^{\ell-1,2}}^2,$$

so with $K = K_2 + CK_1$ the result follows. $\square$

5 More bounds

In section 3 we found bounds on the spinor ψ and the curvature F_A^+ which essentially only depended on the geometry of X . In this section we will obtain bounds on dF_A^+ and A .

Lemma 5.1. *There is a constant C depending only on X and the perturbation parameter η such that for any solution (ψ, A) to the Seiberg-Witten equations we have*

$$\|dF_A^+\|_{L^2}^2 \leq C.$$

Proof. Let ∇^g denote the Levi-Cevita connection. Then we have by the spinor connection property and the Seiberg-Witten equation that

$$\nabla^g F_A^+ = \nabla(\bar{\psi} \otimes \psi - \frac{|\psi|^2}{2}\text{id}) - i\nabla^g \eta_+.$$

By lemma B.1, the anti-symmetrization of the Levi-Civita connection applied to a 2-form is precisely the exterior derivative of the 2-form, so by antisymmetrizing the equation above we obtain

$$dF_A^+ = \nabla(\bar{\psi} \otimes \psi - \frac{|\psi|^2}{2}).$$

Then we obtain

$$dF_A^+ = \nabla(\bar{\psi}) \otimes \psi + \bar{\psi} \otimes \psi - \text{Re}\langle \nabla \psi, \psi \rangle \text{id}.$$

All terms on the right hand side are L^2 -bounded with bounds only depending on the geometry of X and the perturbation parameter, so we obtain a bound C for $\|dF_A^+\|_{L^2}^2$ $\square$

Lemma 5.2. *There is a constant C_1 only depending on X and the perturbation parameter η such that for any solution (ψ, A) to the Seiberg-Witten equations we have*

$$\|F_A^+\|_{L^{1,2}}^2 \leq C_1.$$

Proof. Let $\pi: L^{1,2}(\wedge_+^2 T^*X) \rightarrow H^\perp$ be the orthogonal projection to the orthogonal complement of the space of harmonic self-dual two forms. We may then write $F_A^+ = \pi(F_A^+) + \omega$, with ω self-dual and harmonic. Then Hodge theory gives a constant C' such that

$$\|\pi(F_A^+)\|_{L^{1,2}}^2 \leq C'\|dF_A^+\|_{L^2}^2.$$

We also have an orthogonal projection to self-dual harmonic two forms, so there is a constant $C'' > 0$ such that $\|\omega\|_{L^{1,2}} \leq C''\|F_A^+\|_{L^2}$. Since we have an L^2 -bound on F_A^+ and a L^2 bound on dF_A^+ it follows that the asserted $C_1 > 0$ exists. $\square$

Proposition 5.3. *Let σ be a spin^c structure and let A_0 be a fixed C^∞ connection on the determinant bundle $\det(\sigma)$. Then there exists a constant K_1 depending only on X , the perturbation parameter η and A_0 such that for any solution (ψ, A) to the Seiberg-Witten equations we have a connection $A' = A_0 + \alpha$ gauge equivalent to A with $d^*\alpha = 0$ and $\|\alpha\|_{L^{2,2}}^2 \leq K_1$.*

Proof. This follows from lemma 4.1 combined with the previous lemma. $\square$

6 Compactness of the moduli space

The following theorem is the key to showing compactness of the moduli space.

Theorem 6.1. *Suppose that (ψ, A) is a solution to the Seiberg-Witten equations and that we have fixed gauge so that $A = A_0 + \alpha$ where A_0 is a fixed C^∞ connection on the determinant line bundle, with $d^*\alpha = 0$ and with the projection of α into the harmonic forms contained in a given compact fundamental domain modulo the lattice of harmonic forms with periods in $4\pi i\mathbb{Z}$. For every $\ell \geq 2$ there is a constant $C(\ell)$, depending only on X , A_0 , the perturbation parameter and ℓ such that*

$$\|\alpha\|_{L^{\ell,2}}^2 + \|\psi\|_{L^{\ell,2}}^2 \leq C(\ell).$$

(Here the $L^{\ell,2}$ -norm of the spinor is taken with respect to ∇_{A_0}).

Proof. We have proven that ψ is pointwise bounded and we have seen that α is $L^{2,2}$ bounded. We have $\nabla_A \psi = \nabla_{A_0} \psi + \alpha \psi$, so the $L^{2,2}$ -bound on ψ together with the L^2 bound on ∇_A yield a $L^{1,2}$ bound on ψ .

We will show that ψ is bounded in $L^{3,2}$. By the Dirac equation we have

$$\not{D}_{A_0} \psi = -\alpha \cdot \psi. \tag{1}$$

Since α is $L^{2,2}$ bounded and ψ is C^0 bounded, it follows that $\not{D}_{A_0} \psi$ is L^4 bounded, so by ellipticity of $\not{D}_{A_0}$, the projection to the orthogonal of $\ker(\not{D}_{A_0})$ is $L^{1,4}$ bounded. Since ψ is also L^2 bounded, so is its projection to $\ker(\not{D}_{A_0})$. Since this is finite dimensional, all norms are equivalent, and so the projection is also $L^{1,4}$ bounded. So together these imply a $L^{1,4}$ bound on ψ .

Then using lemma 2.4, we obtain an $L^{1,3}$ bound on $\not{D}_{A_0} \psi$, so arguing in the same way, we obtain a $L^{2,3}$ bound on ψ . Then using lemma 2.3, we obtain a $L^{2,2}$ bound on $\not{D}_{A_0} \psi$ and so once again we obtain a $L^{3,2}$ bound on ψ .

Now, from the curvature equation

$$F_A^+ = \bar{\psi} \otimes \psi - \frac{|\psi|^2}{2} \text{id} - i\eta_+ \tag{2}$$

and lemma 2.2 it follows that F_A^+ is also bounded in $L^{3,2}$. So by the gauge fixing lemma 4.1 α is $L^{4,2}$ bounded. This was the initial step of obtaining a bound $C(\ell)$.

Now suppose by induction that we have for some $\ell \geq 3$ bounds for the $L^{\ell,2}$ -norms of α and ψ . Then from equation 1 and lemma 2.2 it follows that there is a $L^{\ell,2}$ bound on $\not{D}_{A_0} \psi$ and so there is a $L^{\ell+1,2}$ bound on ψ . Then from the curvature equation 2 it follows that there is a $L^{\ell,2}$ -bound on F_A^+ . So by the gauge fixing lemma 4.1, there is a $L^{\ell+1,2}$ bound on α . The result follows by induction. $\square$

Corollary 6.2. *Let (ψ_n, A_n) be any sequence of solutions to the Seiberg-Witten equations. Then after passing to a subsequence, and applying $L^{3,2}$ gauge transformations we can arrange that the (ψ_n, A_n) are C^∞ objects and they converge in the C^∞ topology to a limit (ψ, A) which is also a solution to the Seiberg-Witten equations.*

Proof. By Morrey's theorem we have compact embeddings $L^{\ell,2} \hookrightarrow C^{\ell-3}$. By the theorem we can gauge fix each (ψ_n, A_n) with a $L^{3,2}$ gauge transformation to obtain a sequence of $(\psi_n, A_0 + \alpha_n)$ as in the theorem. For these α_n and ψ_n we have $L^{\ell,2}$ bounds only depending on ℓ , X , a choice of A_0 and the perturbation parameter η . We now apply a diagonal argument to obtain a C^∞ convergent subsequence.

We inductively define subsequences $(\psi_n^{(\ell)}, A_0 + \alpha_n^{(\ell)})$ for all $\ell \geq 3$. For $\ell = 3$, we have a compact embedding $(\psi_n, A_0 + \alpha_n) \in C^0$. Since we have an a priori $L^{3,2}$ bound on ψ_n and α_n , we obtain a convergent subsequence $(\psi_n^{(3)}, A_0 + \alpha_n^{(3)})$. Now, suppose we have defined $(\psi_n^{(\ell)}, A_0 + \alpha_n^{(\ell)})$. Then by the compact embedding $L^{\ell+1,2} \hookrightarrow C^{\ell-2}$ and the $L^{\ell+1,2}$ bound on the ψ_n and α_n , we have a convergent subsequence $(\psi_n^{(\ell+1)}, A_0 + \alpha_n^{(\ell+1)})$ of $(\psi_n^{(\ell)}, A_0 + \alpha_n^{(\ell)})$.

Now, we define $(\mu_n, B_n) = (\psi_n^{(n+3)}, A_0 + \alpha_n^{(n+3)})$. Then we have that (μ_n, B_n) is a subsequence of (ψ_n, A_n) and it is convergent for all C^k . So it is a C^∞ convergent sequence, say $(\mu_n, B_n) \rightarrow (\mu, B)$. Since all (μ_n, B_n) solve the Seiberg-Witten equations, it follows that (μ, B) is also a solution to the Seiberg-Witten equations. $\square$

Corollary 6.3 (Compactness of the moduli space). *The moduli space $\mathcal{M}_\sigma^{k+1}$ is compact.*

Proof. Since Sobolev spaces are separable, they are second countable. Therefore the notions of compactness and sequential compactness coincide. The previous corollary precisely states that the moduli space is sequentially compact. $\square$

References

- [1] A. Behzadan and M. Holst. "Multiplication in Sobolev spaces, revisited". In: *Arkiv för Matematik* 59.2 (Oct. 2021).

A Proper group actions

Definition A.1. Let X be a topological space and let G be a topological group. Assume that $G \curvearrowright X$ continuously. We say that the action is *proper* if the shear map $S: G \times X \rightarrow X \times X, (g, x) \mapsto (x, g \cdot x)$ is proper.

We have the following alternative characterizations.

Proposition A.2. *Let $G \curvearrowright X$ be a continuous group action. Assume that G and X are metrizable. Then the following are equivalent:*

1. *The action is proper.*
2. *For all sequences $(x_n) \subset X$ and $(g_n) \subset G$ such that $x_n \rightarrow x$ and $g_n \cdot x_n \rightarrow y$, there is a convergent subsequence of (g_n) .*
3. *For all compact $K \subset X$, the set $G_K = \{g \in G \mid g \cdot K \cap K \neq \emptyset\}$ is compact.*

Proof. Note that for metrizable space the notions of compactness and sequential compactness coincide. Moreover, since subspaces are metrizable as well, this property is hereditary. We will use this fact in the proof.

(1 $\Rightarrow$ 2) Suppose that the action is proper. Let $(x_n) \subset X$ and $(g_n) \subset G$ be sequences such that $x_n \rightarrow x$ and $g_n \cdot x_n \rightarrow y$. Then for all n we have $S(g_n, x_n) = (x_n, g_n \cdot x_n)$, so by assumption $S(g_n, x_n) \rightarrow (x, y)$. So the set $L = \{S(g_n, x_n) \mid n \in \mathbb{N}\} \cup \{(x, y)\}$ is compact. Since S is proper by assumption, the set $S^{-1}(L) \subset G \times X$ is compact. So $\text{pr}_G(S^{-1}(L)) \subset G$ is compact and $(g_n) \subset \text{pr}_G(S^{-1}(L))$. So by (sequential) compactness, (g_n) has a convergent subsequence.

(2 $\Rightarrow$ 3) Let $K \subset X$ be compact. Let $(g_n) \subset G_K$ be a sequence. Then for each $n \in \mathbb{N}$ we can choose an $x_n \in K$ such that $g_n \cdot x_n \in K$. Then we can extract a subsequence of x_n and g_n which

we denote by the same symbol such that (x_n) and $(g_n \cdot x_n)$ converge, using the compactness of K . Then by 2 we have a convergent subsequence of g_n . So G_K is sequentially compact, and so also compact.

(3 $\Rightarrow$ 1) We have to show that the shear map $S: G \times X \rightarrow X \times X$ is proper. Let $K \subset X \times X$ be compact. Then $K_1 = \text{pr}_1(K)$ and $K_2 = \text{pr}_2(K)$ are compact, hence $L = K_1 \cup K_2$ is compact. Then $L \times L$ is also compact and $K \subset L \times L$. We claim that $S^{-1}(L \times L) \subset G_L \times L$.

Let $(g, x) \in S^{-1}(L \times L)$. Then $(x, g \cdot x) \in L \times L$, so $x \in L$ and $g \cdot x \in L \cap g \cdot L$. So $g \in G_L$. By assumption G_L is compact, so $G_L \times L$ is compact. Finally, by continuity, $S^{-1}(K) \subset S^{-1}(L \times L) \subset G_L \times L$ is a closed subset. So $S^{-1}(K)$ is also compact. So S is proper. $\square$

We have used the following lemma in the proof.

Lemma A.3. *Let X be a topological space. Let $(x_n) \subset X$ be a sequence. Assume that $x_n \rightarrow x$. Then $L = \{x_n \mid n \in \mathbb{N}\} \cup \{x\}$ is compact.*

Proof. Let $\mathcal{U}$ be an open cover of L . Then there is a $U_0 \in \mathcal{U}$ such that $x \in U_0$. Since $x_n \rightarrow x$, there exists an $N \in \mathbb{N}$ such that for $n \geq N$ we have $x_n \in U_0$. Now, for $0 \leq k \leq N-1$, we pick $U_{k+1} \in \mathcal{U}$ such that $x_k \in U_{k+1}$. Then $\{U_0, U_1, \dots, U_N\}$ is a finite subcover. $\square$

Lemma A.4. *Let $G \curvearrowright X$ be a continuous group action. Then the quotient map $q: X \rightarrow X/G$ is open.*

Proof. Let $U \subset X$ be open. Then

$$q^{-1}(q(U)) = \bigcup_{g \in G} g \cdot U$$

which is open. So by the quotient map property, $q(U)$ is open. $\square$

Lemma A.5. *Let $G \curvearrowright X$ be a proper group action. Let $R \subset X \times X$ be the equivalence relation on X induced by the action. Then R is closed as a subset.*

Proof. Let $S: G \times X \rightarrow X \times X$ be the shear map of the action. We claim that $R = S(G \times X)$, which is then closed since the image of a proper map is closed. We have $(x, y) \in R$ iff $x \sim_G y$ iff there exists a $g \in G$ such that $y = g \cdot x$ iff there exists a $g \in G$ such that $S(g, x) = (x, y)$ iff $(x, y) \in S(G \times X)$. So indeed $R = S(G \times X)$. $\square$

Proposition A.6. *Let $G \curvearrowright X$ be a continuous proper group action. Assume that X and G are Hausdorff. Then the quotient space X/G is Hausdorff.*

Proof. Denote the quotient map by $q: X \rightarrow X/G$. Let $q(x), q(y) \in X/G$. Assume that $q(x) \neq q(y)$. Then $(x, y) \notin R$. Since R is closed, there are open neighbourhoods $U, V \subset X$ of x and y respectively, such that $U \times V \cap R = \emptyset$. Then $q(U)$ and $q(V)$ are open neighbourhoods of $q(x)$ and $q(y)$ respectively. Suppose that $q(z) \in q(U) \cap q(V)$. Then there are $g, h \in G$ such that $g \cdot z \in U$ and $h \cdot z \in V$, so $(g \cdot z, h \cdot z) \in U \times V$ and $(g \cdot z, h \cdot z) \in R$, since $g \cdot z = gh^{-1} \cdot (h \cdot z)$. This is a contradiction, so $q(U) \cap q(V) = \emptyset$. So X/G is Hausdorff. $\square$

B Differential geometry identities

Lemma B.1. *Let M be a manifold and let ∇ be a torsion-free connection on TM . Let Alt denote the antisymmetrization map. Then we have for $\omega \in \Omega^k(M)$*

$$d\omega = \text{Alt}(\nabla\omega),$$

(where ∇ is the induced connection on $\wedge^k T^*M$).

Proof. It suffices to check this on 1-forms, since both sides are anti-derivations. So let $\omega \in \Omega^1(M)$ and $X, Y \in \mathfrak{X}(M)$. Then we have

$$\begin{aligned}
 \text{Alt}(\nabla\omega)(X, Y) &= (\nabla_X\omega)(Y) - (\nabla_Y\omega)(X) \\
 &= X(\omega(Y)) - \omega(\nabla_X Y) - Y(\omega(X)) + \omega(\nabla_Y X) \\
 &= X(\omega(Y)) - Y(\omega(X)) - \omega(\nabla_X Y - \nabla_Y X) \\
 &= X(\omega(Y)) - Y(\omega(X)) - \omega([X, Y]) = d\omega(X, Y).
 \end{aligned}$$

□